Legal-Aid RAG MLOps Pipeline

Milestone 1: Model, API, MLflow Tracking, and GenAI Evaluation

A reproducible legal-inheritance retrieval-augmented generation system with validated corpus preparation, eight retrieval profiles, MLflow GenAI datasets and judges, controlled inference ablations, FastAPI serving, and a local Gradio interface.

Team member	ERP ID	Primary responsibility
Karam Hussain	27857	Modeling, evaluation, MLflow experiment design, and champion selection
Aadesh Panjwani	29303	Data pipeline, retrieval implementation, API, and generation metadata logging
Muhammad Rayyan Khan	28410	Testing, manual verification, reproducibility, and quality assurance

Submission branch: **8-m1-submission**

Report date: 5 November 2026

Repository: github.com/karnage4/MLOps-for-Legal-Aid-RAG-pipeline

Executive summary

This milestone converts the previous notebook and application prototype into a reproducible RAG delivery pipeline. The original architecture is preserved: local BM25 retrieval is combined with Pinecone semantic retrieval, candidates are fused and reranked, Hugging Face remains the primary generation provider, and Groq is used as a clean fallback. MLflow records conventional experiments, nested query-retrieval-generation traces, a managed evaluation dataset, versioned GenAI judges, and controlled comparisons.

The traditional supervised-training requirement is explicitly treated as exempt because the project uses pretrained embedding, reranking, and generation models. In its place, the system performs inference evaluation across retrieval parameters, prompts, models, and temperatures, matching the instructor's RAG-specific guidance.

Final status: all eight retrieval configurations and all four controlled inference-ablation configurations completed successfully. The managed six-judge comparison selected fixed-hybrid-top5, while the generation ablation selected structured_legal_v2 with Qwen at temperature 0.1.

Milestone compliance matrix

Requirement	Status	Evidence
README pitch, architecture, quick start, dataset	Complete	README includes reproducible setup, architecture, corpus details, and public Drive access
Separate data preparation and validation	Complete	prepare_bm25.py validates fields, empty values, ranges, duplicates, strategy, and persisted alignment
Training and model artifact	RAG exemption	Inference evaluation and clearly named prepared/evaluation artifacts replace unrelated supervised training
At least three MLflow experiment runs	Complete	8 retrieval runs, 8 managed GenAI runs, and 4 controlled generation ablations
FastAPI prediction service	Complete	/health, /predict, Pydantic validation, component cache, and /docs
Unit tests	Complete	Preparation, loading, fallback, tracing, evaluation, API, and Gradio coverage
Contribution document	Complete	Names, ERP IDs, task division, and detailed responsibilities

1. System architecture and design decisions

The pipeline separates preparation, indexing, retrieval, generation, evaluation, and serving. Fixed and recursive chunk stores are isolated through matching BM25 artifacts and Pinecone namespaces. Retrieval can run dense semantic search only or a hybrid path that combines BM25 and dense candidates using reciprocal-rank fusion before cross-encoder reranking.

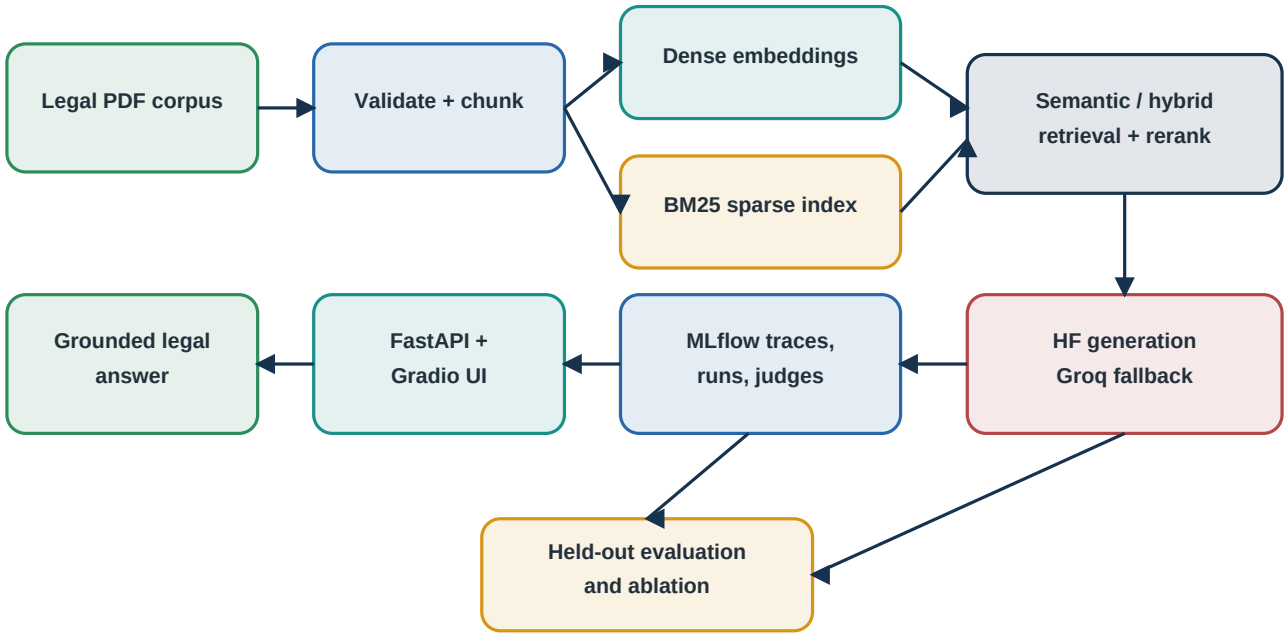


Figure 1. End-to-end architecture of the final Milestone 1 RAG pipeline.

Key architectural guarantees

- Retrieval behavior remains compatible with the original project; configurable profiles are layered around the same semantic and hybrid components.

- Every application query produces one MLflow CHAIN trace with RETRIEVER and LLM child spans, preserving parent-child nesting.

- Hugging Face is attempted first in production auto mode. Groq uses GROQ_API_KEY from .env only when the primary provider is unavailable.

- Evaluation runs can request direct Groq generation for controlled model comparisons without changing the production fallback default.

- Secrets, full prompts, and full retrieved documents are not returned by the public API response contract.

Technology stack

Layer	Technology	Purpose
Preparation	Python, PDF extraction, validation	Create fixed and recursive chunks and aligned BM25 corpora
Dense retrieval	Sentence Transformers + Pinecone	Semantic candidate retrieval with isolated namespaces
Sparse retrieval	BM25	Exact-term and legal phrase matching
Reranking	Cross-encoder MiniLM	Improve final candidate ordering
Generation	Hugging Face + Groq	Primary/fallback answer generation
MLOps	MLflow 3.17.x	Runs, nested traces, datasets, judges, assessments, and artifacts
Serving	FastAPI + Gradio	Validated REST API and interactive local interface

Operational modes

Mode	Behavior
Production auto	Hugging Face first; bounded Groq fallback if credits, provider availability, or transient requests fail
Controlled evaluation	Direct requested Groq model to isolate prompt, model, and temperature effects
Semantic retrieval	Pinecone dense candidates only, followed by the shared response pipeline
Hybrid retrieval	BM25 plus Pinecone candidates, reciprocal-rank fusion, and cross-encoder reranking

Milestone design rationale

The implementation avoids introducing an unrelated classifier simply to satisfy a traditional training rubric. Instead, it treats retrieval configuration and generation behavior as the model system under evaluation, preserving architectural validity while producing reproducible MLflow evidence.

2. Dataset access, preparation, and validation

The source corpus contains legal inheritance material grouped into five categories: core statutory laws, Islamic fiqh-based material, research and legal analysis, government and legal reports, and case law. The folder is publicly viewable for the instructor and teaching assistant. Raw PDFs remain outside Git to keep source distribution separate from code and to avoid repository bloat.

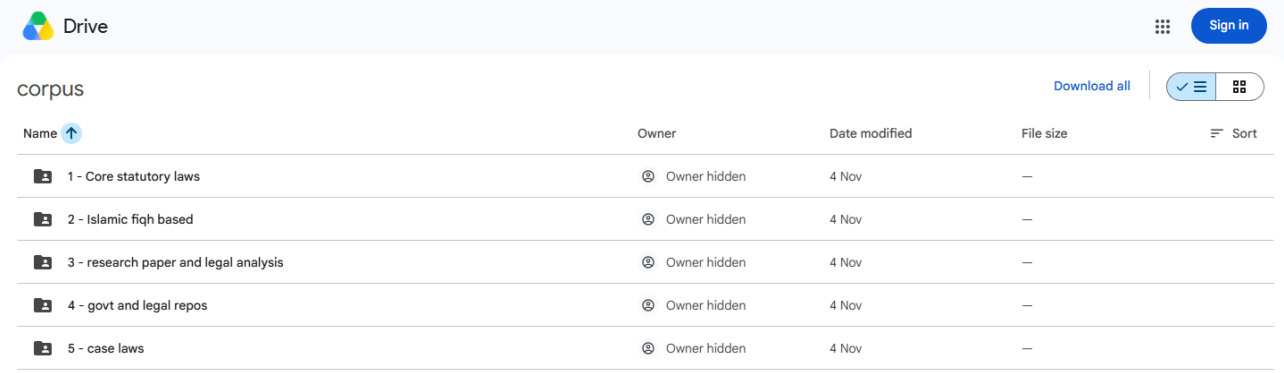


Figure 2. Anonymous Google Drive view showing the five public corpus folders.

Public corpus: Google Drive folder. Anonymous HTTP and visual access were verified on 5 November 2026.

Preparation outputs

Item	Verified result
Input files	50 supplied PDFs across five folders
Accepted files	46 PDFs with sufficient machine-readable content
Skipped files	4 PDFs below the minimum extractable-text threshold
Recursive chunks	3,593 validated chunks
Fixed chunks	883 validated chunks
Sparse artifacts	Strategy-specific BM25 JSON corpora
Dense storage	Recursive default namespace and isolated fixed namespace in Pinecone

Validation controls

- Reject malformed JSON and non-list chunk collections.
- Require non-empty id, text, and source fields.
- Reject negative or non-integer chunk indexes.
- Reject duplicate IDs and chunking-strategy mismatches.
- Persist, reload, and revalidate BM25 alignment before downstream use.

3. MLflow integration and experiment lineage

MLflow is configured through a shared module and defaults to a repository-local SQLite tracking store. The same experiment captures smoke tests, offline comparisons, nested RAG traces, managed datasets, judge definitions, and controlled evaluation artifacts. Optional MLFLOW_TRACKING_URI and MLFLOW_EXPERIMENT_ID variables support migration to a shared server.

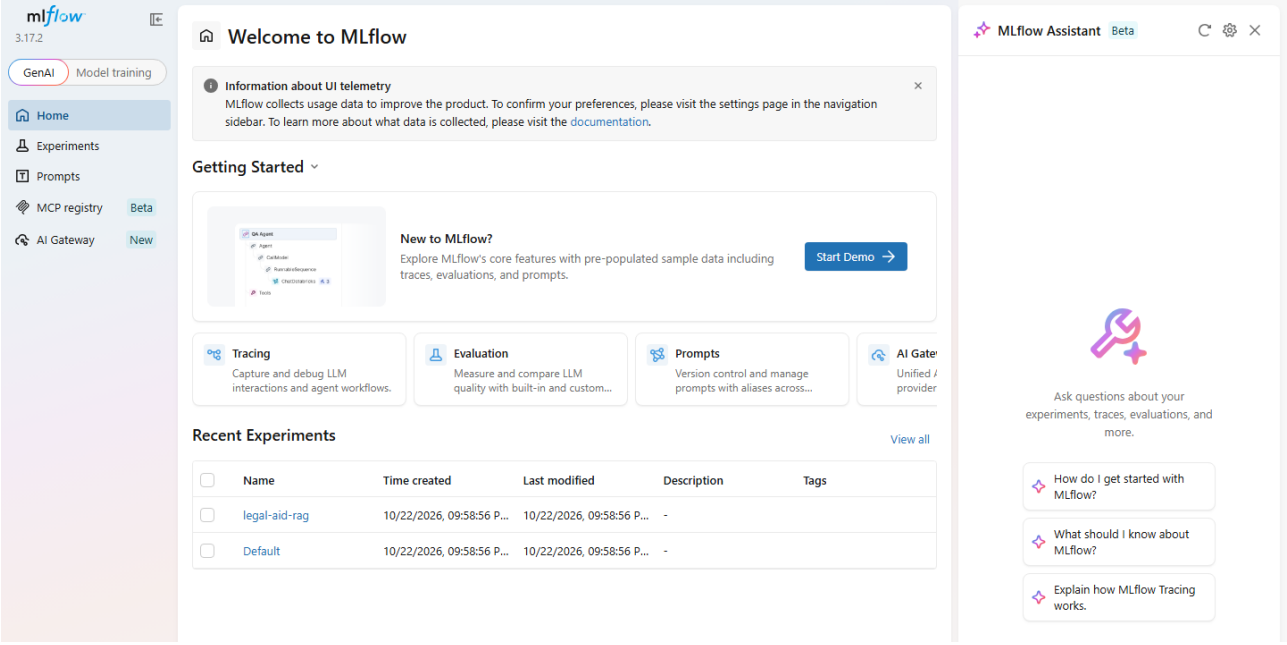


Figure 3. Repository-local MLflow 3.17.2 server showing the legal-aid-rag experiment.

Nested trace contract

Span	Type	Recorded evidence
legal_aid_rag_query	CHAIN	Query, retrieval profile, temperature, requested provider/model, final response metadata
hybrid_legal_retrieval	RETRIEVER	Strategy, namespace, top-K, sparse/dense candidate counts, reranking, latency
legal_generation	LLM	Provider attempts, actual provider/model, fallback flag, latency, sanitized outcomes

Managed GenAI assets

- Dataset legal-aid-rag-held-out-v1 contains three idempotently registered held-out questions and human-reviewed expected facts.

- Six pinned judges assess query relevance, retrieval relevance, retrieval groundedness, correctness, retrieval sufficiency, and legal-answer quality.

- Automatic trace evaluation remains off; the registered judges are invoked explicitly on the controlled dataset to prevent unbounded provider usage.

- Partial or timed-out runs are marked invalid and excluded from ranking until every expected prediction and judge value is present.

4. Retrieval-parameter evaluation

The first experiment suite evaluates the exact Cartesian product of two chunking strategies, two retrieval modes, and two top-K values. Every profile uses the same three questions, the same generation temperature (0.2), and the same evaluation formulas. All 24 query executions completed successfully.

Configuration	Faithfulness	Relevancy	Avg total ms	Run ID
recursive-semantic-top3	0.8889	0.7717	17,208.0	db8bf52010044b47a262af5e8add6dda
recursive-semantic-top5	0.7556	0.4809	6,851.3	de06c99b5b9d41d89333459657065f59
recursive-hybrid-top3	0.9667	0.7860	11,799.0	07be884a244d41c99f5f8540611a07f1
recursive-hybrid-top5	0.7778	0.7723	9,374.7	6d8ff2c40dce4ba48aa2bc77f98d95b3
fixed-semantic-top3	1.0000	0.7228	10,589.7	73841e75f54b4a968339b8b63943d5d4
fixed-semantic-top5	0.7963	0.7501	16,014.0	cc8313f367b7424eba49bf45328803f9
fixed-hybrid-top3	1.0000	0.7700	17,530.0	ba6464e78f014a6db1d21af5c39d508f
fixed-hybrid-top5	0.6667	0.7402	15,932.7	84ec54970978451fb33ace36af25c730

Legacy evaluator winner: fixed-hybrid-top3 (run ba6464e78f014a6db1d21af5c39d508f). It tied for highest faithfulness at 1.0000 and had higher relevancy than the other 1.0000-faithfulness profile.

Interpretation

The results show that retrieval quality is not determined by one universal setting. Recursive hybrid top-3 produced the highest mean relevancy (0.7860), while fixed configurations achieved the highest faithfulness. This justified keeping chunking, mode, and top-K configurable rather than hard-coding a single profile throughout the application.

5. Managed six-judge GenAI comparison

The second evaluation suite reuses the same eight profiles but applies the MLflow-managed dataset, reviewed expectations, and all six pinned judges. Ranking prioritizes correctness, retrieval sufficiency, legal-answer quality, grounding, relevance, and latency under strict completeness rules.

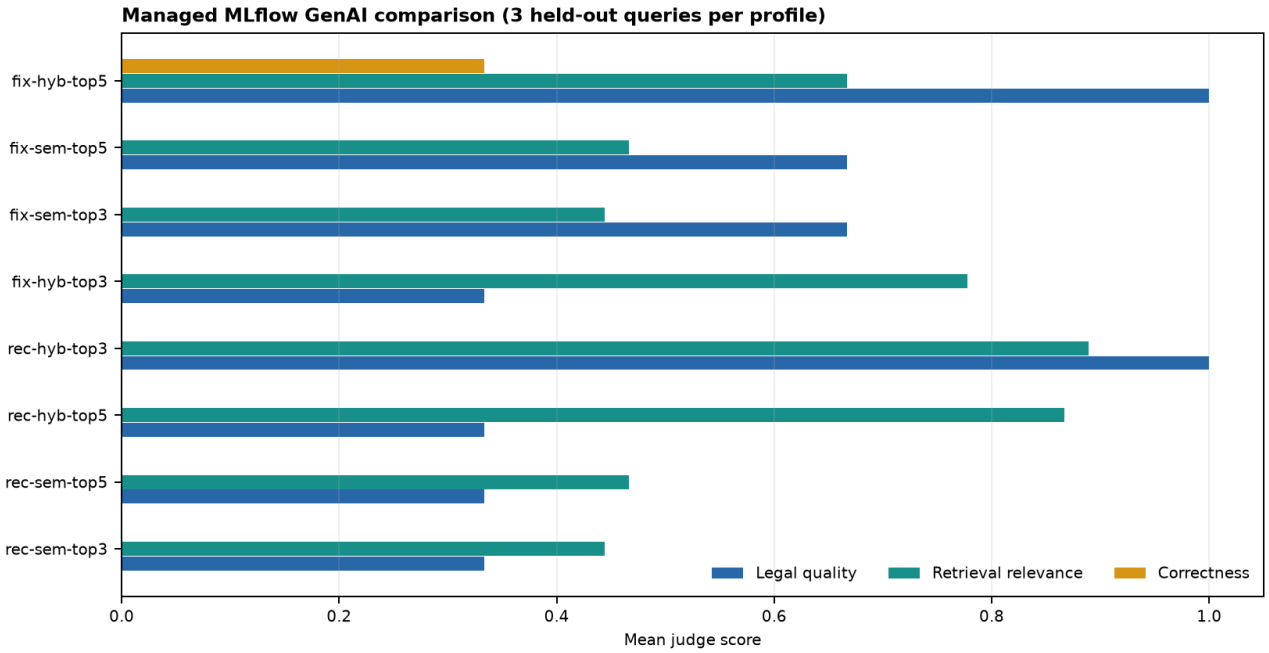


Figure 4. Quality metrics across all eight complete managed GenAI profiles.

Profile	Correct	Sufficient	Legal quality	Grounded	Relevant	Run ID
fixed-hybrid-top5	0.333	0.333	1.000	0.667	0.667	e6034e3cda2f4582ab5fac02132b4687
fixed-semantic-top5	0.000	0.333	0.667	1.000	0.467	68e216b95fc3455bad64fd8f516c2902
fixed-semantic-top3	0.000	0.333	0.667	1.000	0.444	91c3ff85269a47ad8a1131b5e28815dc
fixed-hybrid-top3	0.000	0.333	0.333	1.000	0.778	9fcc1af514b046078f04bc2202c5f60a
recursive-hybrid-top3	0.000	0.000	1.000	1.000	0.889	98e7838d135f43fcb830c1d824e9581a
recursive-hybrid-top5	0.000	0.000	0.333	1.000	0.867	63fe861d46e740a0893c7d13041152ba
recursive-semantic-top5	0.000	0.000	0.333	1.000	0.467	a130d951e9ce446d9238848c24dd8986
recursive-semantic-top3	0.000	0.000	0.333	0.333	0.444	5e585a134b844a0a8e50ae4693cee229

Managed GenAI winner: fixed-hybrid-top5 (run e6034e3cda2f4582ab5fac02132b4687). It was the only profile to reach correctness 0.333 while also achieving legal-answer quality 1.000 and query relevance 1.000.

Why the two retrieval winners differ

The legacy suite selects fixed-hybrid-top3 using faithfulness, relevancy, and latency. The managed GenAI suite selects fixed-hybrid-top5 using a broader rubric that includes reviewed expected facts, legal framing, sufficiency, and correctness. The results are therefore complementary rather than contradictory. The latter profile is frozen for the controlled generation ablation because it won the richer six-judge comparison.

Pinned judge rubric

Judge	Purpose	Expected facts
relevance_to_query	Does the answer address the question?	No
retrieval_relevance	Are retrieved chunks relevant to the query?	No
retrieval_groundedness	Is the answer supported by retrieved evidence?	No
legal_answer_quality	Does the answer communicate conditions, uncertainty, and responsible framing?	No
correctness	Does the answer cover the approved expected facts?	Yes
retrieval_sufficiency	Can retrieval support the approved expected facts?	Yes

6. Prompt, model, parameter, and temperature ablation

The instructor's RAG-specific update requires inference comparisons across prompts, models, parameters, and temperature. Parameter variation is covered by the eight retrieval profiles. A separate four-run ablation freezes retrieval to fixed-hybrid-top5 and changes exactly one generation variable at a time. Direct Groq mode is used for controlled model identity while production auto mode remains Hugging Face first.

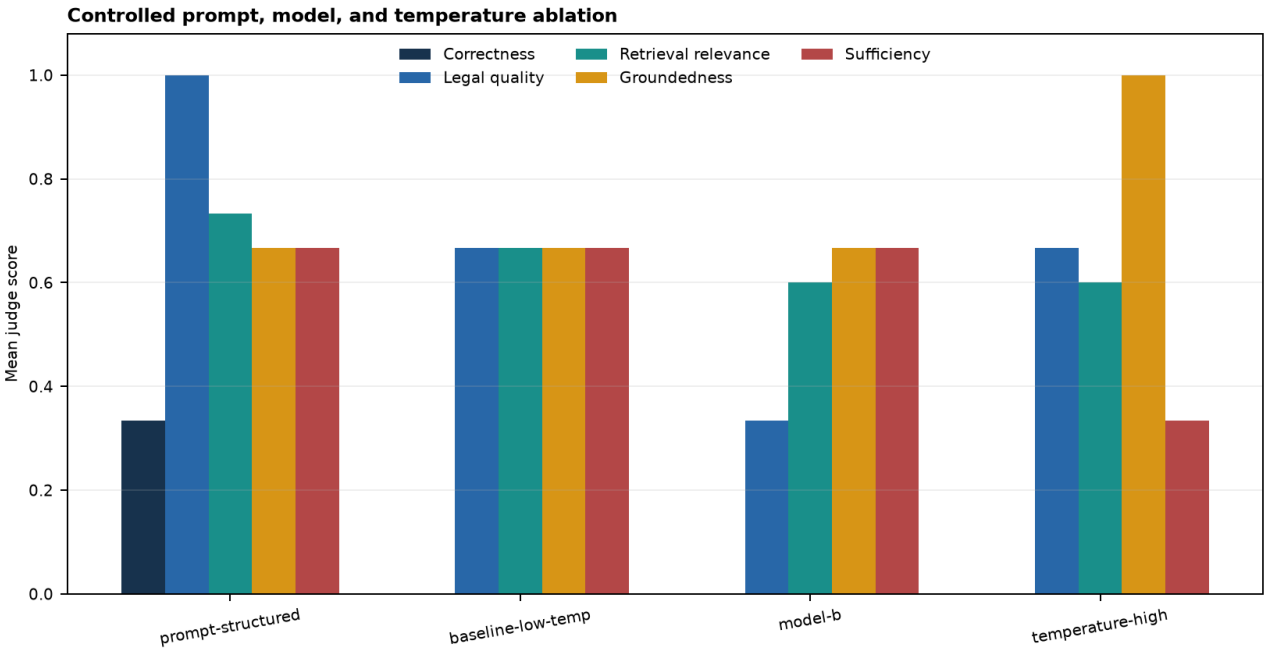


Figure 5. Controlled prompt, model, and temperature results on the same dataset and judge lineage.

Run	Prompt	Model	Temp	Correct	Legal	Relevance	Latency ms
baseline-low-temp	strict_grounded_v1	Qwen 3.8-27B	0.1	0.000	0.667	0.667	28,756
prompt-structured	structured_legal_v2	Qwen 3.8-27B	0.1	0.333	1.000	0.733	21,272
model-b	strict_grounded_v1	GPT-OSS-120B	0.1	0.000	0.333	0.600	20,145
temperature-high	strict_grounded_v1	Qwen 3.8-27B	0.7	0.000	0.667	0.600	16,698

Controlled findings

- Prompt: structured_legal_v2 improved correctness from 0.000 to 0.333, legal quality from 0.667 to 1.000, and retrieval relevance from 0.667 to 0.733.

- Model: GPT-OSS-120B was faster than Qwen in this sample but reduced legal quality to 0.333 and retrieval relevance to 0.600.

- Temperature: raising 0.1 to 0.7 reduced retrieval sufficiency from 0.667 to 0.333 and relevance from 0.667 to 0.600, while groundedness increased to 1.000.

- Selected generation configuration: structured_legal_v2 with qwen/qwen3.8-27b at temperature 0.1.

These scores are milestone evidence from three held-out examples, not statistically generalizable claims. Expanding the reviewed dataset is the most important next evaluation improvement.

6.1 Exact prompt variants used

The controlled prompt comparison changed only the system instruction. Both variants received the same retrieved context and user question through the shared user-message template shown below.

strict_grounded_v1 - baseline system prompt

```
You are a knowledgeable legal assistant specializing in inheritance law.
Your role is to answer questions based strictly on the provided legal context.

Rules you must follow:
- Answer ONLY from the provided context. Do not use outside knowledge.
- If the context does not contain enough information, say:
  "The provided documents do not contain sufficient information to answer this question."
- Cite the source document name when referencing specific rules or sections.
- Be precise and use correct legal terminology.
- Do not give personal legal advice - remind users to consult a qualified lawyer for their spec
ific situation.
```

structured_legal_v2 - structured system prompt

```
You are an evidence-grounded inheritance-law research assistant.

Use only the supplied legal context. Produce a concise answer with these sections:
1. Direct answer
2. Applicable rule and conditions
3. Sources
4. Important limitation

Requirements:
- Cite the source document for every material legal rule.
- Clearly distinguish general rules from conditional or disputed rules.
- If the context is insufficient, explicitly say what cannot be established.
- Never invent statutes, cases, shares, jurisdictions, or facts.
- Give general legal information only and recommend a qualified lawyer for advice.
```

Common user-message template

```
Using the legal context provided below, answer the following question.

--- LEGAL CONTEXT ---
{retrieved context}
--- END CONTEXT ---

Question: {query}

Answer:
```

7. FastAPI prediction service

The service exposes /health and /predict through FastAPI. Pydantic rejects blank queries, unknown fields, top-K values outside 3-10, and temperatures outside 0-1. Retriever components are loaded once and cached at application startup. Errors are mapped to compact HTTP responses without returning secrets, full prompts, or complete retrieved documents.

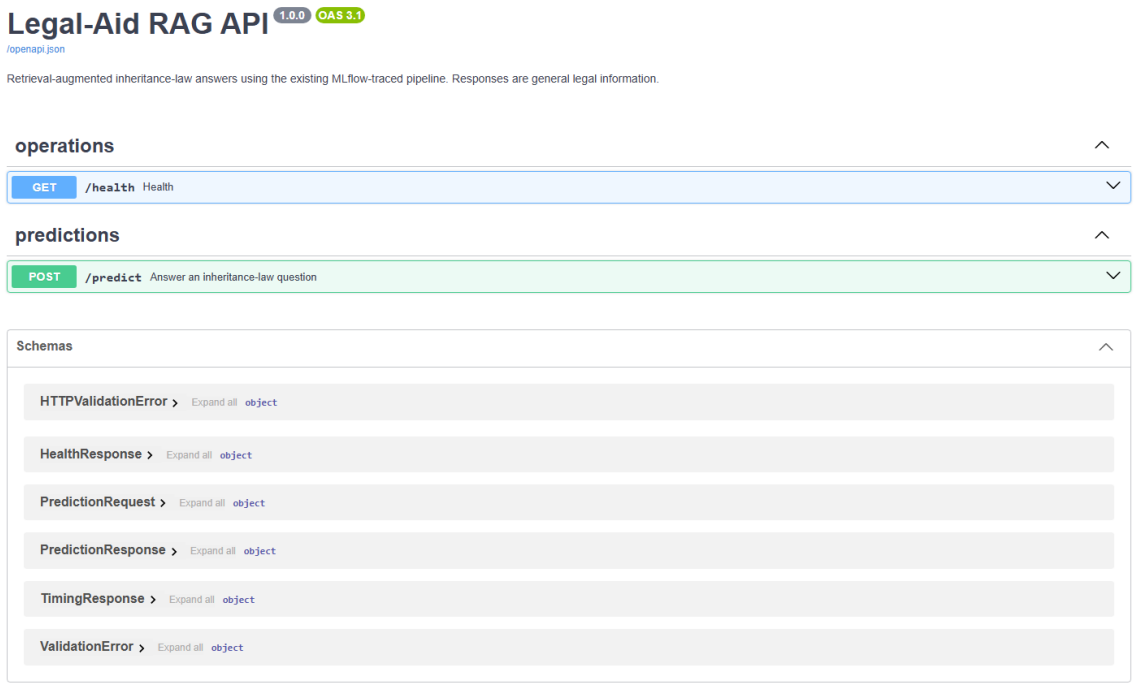


Figure 6. Auto-generated FastAPI documentation showing health, prediction, and validated schemas.

Request example

```
POST /predict
{
  "query": "Who inherits without a will?",
  "chunking_strategy": "fixed",
  "retrieval_mode": "hybrid",
  "top_k": 5,
  "temperature": 0.2
}
```

Response evidence

The compact response includes the answer, actual provider and model, fallback flag, retrieval profile, source identifiers, trace ID, and retrieval/generation/total timing. Manual acceptance confirmed both /health and /predict through /docs.

8. Local Gradio interface

Miras provides a local interface similar to the earlier Hugging Face application while using the same traced RAG boundary. Users can select recursive or fixed chunks, semantic or hybrid retrieval, top-K from 3 to 10, generation temperature from 0 to 1, and whether to display retrieved context.

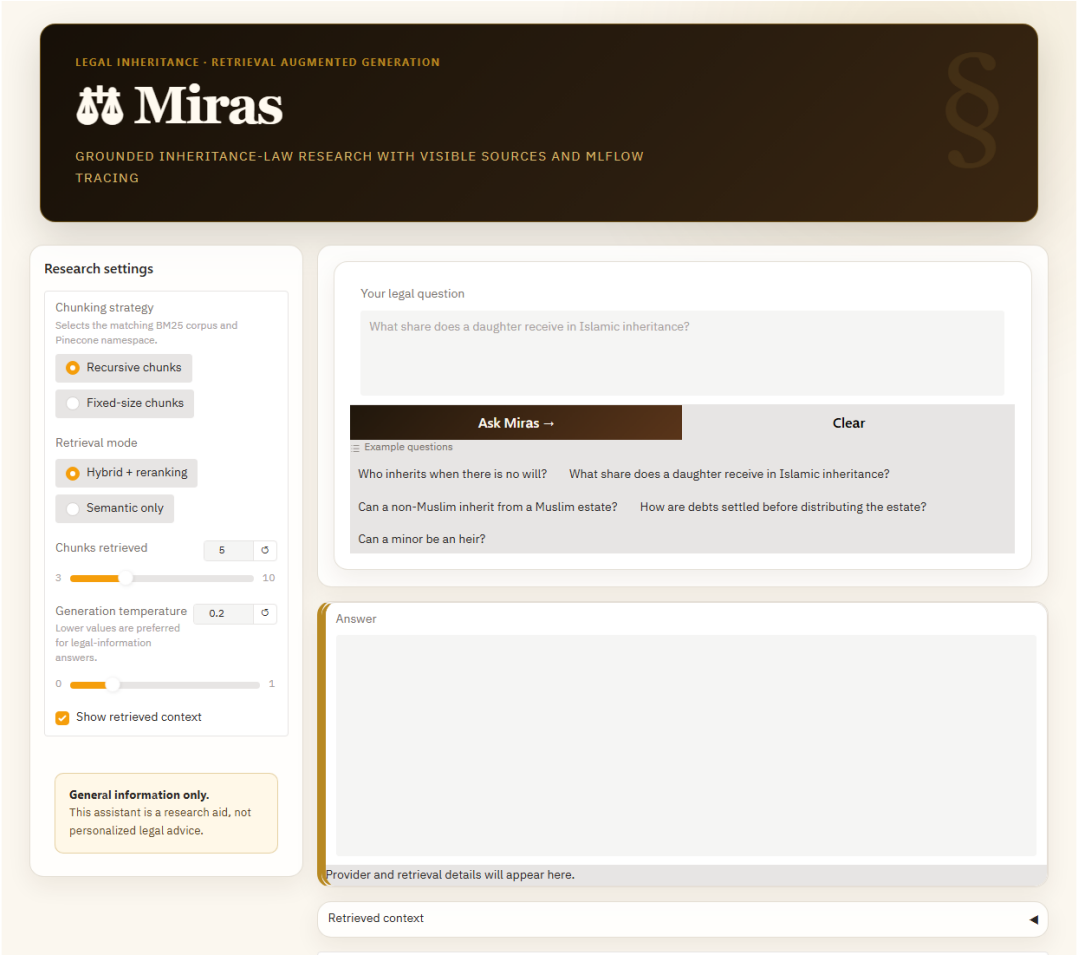


Figure 7. Miras interface showing chunking, retrieval, top-K, temperature, and context controls.

The interface displays an explicit general-information disclaimer and surfaces provider, retrieval, and timing metadata after prediction. Unit, HTTP, live RAG, fallback, and nested-trace verification have passed.

9. Testing, validation, and operational quality

Verification status: the full repository regression completed with 91 passing tests and one pre-existing MLflow SQLAlchemy deprecation warning. The final focused evaluation regression completed with 14 passing tests after the high-temperature run.

Test area	Representative coverage
Preparation	Missing values, invalid ranges, duplicate IDs, strategy mismatch, persisted alignment
Embedding/index	Batch preparation, namespace routing, loader inputs, failure handling
Retrieval	Semantic/hybrid execution, top-K profiles, namespace guards, live smoke checks
Generation	HF-first order, Groq fallback, retries, provider metadata, secret exclusion
Tracing	Single root chain and nested retriever/generation parent IDs
Evaluation	Aggregation, incomplete judge output, strict reuse, best-run selection, eight-profile matrix
FastAPI	Cached loading, request validation, success response, provider failures, /health
Gradio	Input mapping, chunk selector, response formatting, HTTP startup
Inference ablation	Config isolation, lineage matching, partial-run exclusion, ranking and deltas

Quality safeguards

- .env, local MLflow databases, generated outputs, and raw corpus files are excluded from version control.
- Tracked diffs were scanned for credential-shaped values before submission commits.
- Judge and provider failures use bounded retries and explicit partial-run status rather than silent score omission.
- The replacement Groq key remained local; no API key appears in commits or report evidence.
- A transient Pinecone DNS failure was diagnosed at the trace span and excluded before the successful retry.

10. Reproducibility and execution guide

The following sequence reproduces setup, credential-free tests, local services, preparation, and controlled evaluations. Cloud-backed steps require the three keys documented in .env.example.

```
git clone https://github.com/karnage4/MLOps-for-Legal-Aid-RAG-pipeline.git
cd MLOps-for-Legal-Aid-RAG-pipeline
python -m venv .venv
.venv\Scripts\Activate.ps1
python -m pip install -r requirements.txt
Copy-Item .env.example .env
pytest -q
uvicorn src.app:app --host 127.0.0.1 --port 8000
python gradio_app.py
```

Evaluation commands

```
python -m src.prepare_bm25 --strategy fixed
python -m src.evaluation
python -m src.genai_dataset --creator "Karam Hussain"
python -m src.genai_expectations --reviewer "Karam Hussain"
python -m src.genai_judges
python -m src.genai_comparison
python -m src.inference_ablation
```

Required environment variables

Variable	Purpose
HF_API_TOKEN	Primary Hugging Face generation provider
PINECONE_API_KEY	Dense retrieval from the configured Pinecone index
GROQ_API_KEY	Generation fallback and controlled direct-Groq evaluations
MLFLOW_TRACKING_URI	Optional override for a shared MLflow server
GROQ_JUDGE_MODEL	Optional judge model override; verified default is openai/gpt-oss-20b

11. Team contributions

Member	ERP	Responsibility	Verified deliverables
Karam Hussain	27857	Modeling and evaluation	Retrieval experiments, held-out dataset and judges, MLflow tracking and nested tracing, eight-profile comparison, prompt/model/temperature ablation, champion selection, findings documentation, and technical integration.
Aadesh Panjwani	29303	Pipeline and API	Data preparation and validation, fixed/recursive artifacts, BM25 and Pinecone support, semantic/hybrid retrieval, FastAPI endpoints and docs, generation metadata logging, and architecture documentation.
Muhammad Rayyan Khan	28410	Quality and operations	Unit and integration tests, focused/full pytest execution, result and warning review, MLflow trace/judge/artifact verification, manual FastAPI and Gradio validation, reproducibility, and final QA.

Collaboration model

Responsibilities are separated by ownership but verified collaboratively. Modeling decisions are supported by pipeline instrumentation; pipeline changes are protected by automated and manual QA; and final selections are backed by tracked run IDs and reproducible artifacts rather than undocumented judgment.

12. Limitations, risks, and next steps

Current limitation or risk	Mitigation / next action
Only three human-reviewed evaluation questions	Expand the managed dataset across more inheritance scenarios and document strata
LLM judges can vary and are provider-quota sensitive	Retain pinned versions, strict completeness, retries, and add human review on larger samples
Hugging Face credits/provider availability caused fallback	Keep provider abstraction and monitor primary/fallback success rates
Pinecone and external APIs require network connectivity	Add availability monitoring, clearer retry budgets, and optional offline retrieval fixtures
Legal answers are informational, not legal advice	Preserve disclaimers, source visibility, uncertainty language, and qualified-lawyer guidance
Local MLflow SQLite is single-user oriented	Move to a shared tracking server and artifact store for team deployment
No hosted Space in this milestone	Defer deployment until platform choice and tracking topology are approved

Recommended next milestone

- Increase dataset coverage and add reviewer provenance for every expectation.
- Package the API and UI with a production-ready deployment configuration.
- Run automated CI tests and artifact validation on pull requests.
- Deploy shared MLflow tracking with durable artifacts and access controls.
- Add monitoring for provider fallback rate, latency, retrieval drift, and answer-quality regressions.

13. Evidence index and conclusion

Key run IDs

Evidence	Selected configuration	MLflow run ID
Legacy retrieval evaluator	fixed-hybrid-top3	ba6464e78f014a6db1d21af5c39d508f
Managed six-judge comparison	fixed-hybrid-top5	e6034e3cda2f4582ab5fac02132b4687
Ablation baseline	strict prompt, Qwen, temp 0.1	f8eb6177b0a44c2e91cbd27926ca3efd
Selected structured prompt	structured prompt, Qwen, temp 0.1	1528a96621d54a93819f01eab6b600a0
Model comparison	strict prompt, GPT-OSS-120B, temp 0.1	6db207d1847d4d96af0f42905fcf1bf1
Temperature comparison	strict prompt, Qwen, temp 0.7	caf08bcd4194455f9efc030ffc2a9634

Tracked evidence

- artifacts/rag_evaluation_summary.json - complete legacy eight-profile comparison.

- artifacts/genai_evaluation_comparison.json - complete managed six-judge ranking.

- artifacts/inference_ablation/comparison.json - complete prompt/model/temperature comparison.

- README.md - setup, architecture, dataset access, commands, evaluation results, and training exemption.

- MLOPS_LOG.md - chronological implementation and manual-verification record.

- CONTRIBUTION.md - final member-to-task allocation.

Conclusion

Milestone 1 is complete under the RAG-specific evaluation interpretation. The project now has a validated and reproducible corpus pipeline, explicit retrieval alternatives, tracked experiments, nested GenAI traces, a reusable dataset with versioned judges, complete inference comparisons, a served prediction API, a local interactive interface, documented tests, and auditable team contributions. The report evidence also makes limitations visible: the current held-out sample is intentionally small, external providers remain operational dependencies, and deployment is deferred. Within those boundaries, the repository provides a clean foundation for the next MLOps milestone.

Repository: MLOps-for-Legal-Aid-RAG-pipeline
Submission branch: **8-m1-submission**
Public corpus: Google Drive corpus folder